J Chain-of-thought completions by GPT-4

GPT-4 reaches human-level performance with CoT prompting. In this section we look at a few completions to help illuminate how CoT-prompting improves implicature resolution.

Specifically, we look at some of the examples for which GPT-4 5-shot gets 0% accuracy (i.e. wrong for all templates), and GPT-4 5-shot CoT gets 100% accuracy (i.e. right for all templates). This is a set of 10 examples. We only look at the first template, but all templates get a similar performance with CoT.

Utterance: 'Do you sell calculators?'
Response: 'We are all sold out. We wont probably have any more in for a few weeks.'
Implicature: Yes. But we dont have any calculators to sell now.

This is an example that GPT-4 gets wrong for all templates 5-shot, probably because the response has a negative sentiment, and also it means they do not sell calculators right now. However, the implicature is yes, because they do sell them normally. The CoT completion is:

CoT completion: 'Alice says they are all sold out and won't have any more in for a few weeks. Alice's response must be relevant to Bob's question because Alice is a cooperative conversational partner. Alice must be implying that they do sell calculators, but they currently don't have any in stock. The answer to Bob's question is yes. Answer: yes'

This example is similar to the one given in the main text in Section 4, and there are more like these in the dataset: the sentiment is negative, and an interpretation can be negative, but strictly the implicature is positive (yes).

Utterance: 'Do you know where he is?'
Response: 'I'll find him.'
Implicature: No.

GPT-4 gets this wrong for all templates 5-shot, and right for all of them with CoT:

CoT completion: 'Alice says she will find him. Alice's response must be relevant to Bob's question because Alice is a cooperative conversational partner. If Alice knew where he is, she would have provided that information directly. Instead, she says she will find him, implying that she does not currently know his location. The answer to Bob's question 'Do you know where he is?' must therefore be no. Answer: no'

It's unclear why GPT-4 gets this one wrong 5-shot (all humans get this one right), but the CoT is very relevant and helpful.

Utterance: 'It seems Cathy is not happy at all with her new room mate. Did you ask her why?'
Response: 'Believe me. I tried to find out. But I simply could not pin it down.'
Implicature: Yes. But Cathy did not give specific reasons for her feelings.

Again an example where the sentiment of the response seems to hint at no. Of 5 human annotators, also only 2 interpret this response as "yes". With a CoT, GPT-4 gets it right six times:

CoT completion: 'Alice says she tried to find out but couldn't pin it down. Alice's response must be relevant to Bob's question because Alice is a cooperative conversational partner. Alice must mean that she attempted to ask Cathy why she is unhappy with her new roommate, but was unable to get a clear answer. The answer to Bob's question 'Did you ask her why?' must therefore be yes. Answer: yes'

A helpful reasoning trace. All CoT completions by the models we have run CoT on are available in the GitHub: `https://github.com/LauraRuis/do-pigs-fly`.

K Additional results

K.1 Contrastive experiment

In this section we reframe the implicature resolution task to a contrastive one, allowing the model to contrast the coherent to the incoherent sentence in a single prompt.

Contrastive task. In the ranking task the model is required to assign higher likelihood to the coherent utterance than the incoherent one ($p_\theta(\mathbf{x}) > p_\theta(\hat{\mathbf{x}})$). In assigning a likelihood to $\mathbf{x}$, the model has no knowledge of $\hat{\mathbf{x}}$, and vice-versa. We hypothesize that the task might become easier if we reformulate it as a contrastive task. Consider the following prompt $\mathbf{p}$.

> Which of the following sentences is coherent:
>
> A: Esther asked "Can you come to my party on Friday?" and Juan responded "I have to work", which means no.
>
> B: Esther asked "Can you come to my party on Friday?" and Juan responded "I have to work", which means yes.
>
> Answer:

We can now evaluate the models' ability to understand which is the coherent sentence by evaluating whether it assigns $p_\theta(A \mid \mathbf{p}) > p_\theta(B \mid \mathbf{p})$. Note that this can again be framed in a ranking task of assigning a higher likelihood to the coherent prompt. If we finish the above prompt $\mathbf{p}$ by adding "A" to make a coherent prompt $\mathbf{x}$ and "B" to make an incoherent prompt $\hat{\mathbf{x}}$ we can again formulate the task by $p_\theta(\mathbf{x}) > p_\theta(\hat{\mathbf{x}})$. The difference is that within both the coherent and the incoherent prompt, the model can contrast the coherent and incoherent utterance to each other. We randomise the assignment of A and B to the utterances.

We do a small experiment with the contrastive task with one of the best performing models overall, OpenAI's text-davinci-002, for $k = \{0, 1, 5\}$. We use two prompt templates and for each template try three different multiple choice answers: A and B like above, one and two, or the full text of the answer. For the last option the coherent prompt $\mathbf{x}$ would look as follows:

> Which of the following sentences is coherent:
>
> A: Esther asked "Can you come to my party on Friday?" and Juan responded "I have to work", which means no.
>
> B: Esther asked "Can you come to my party on Friday?" and Juan responded "I have to work", which means yes.
>
> Answer: Esther asked "Can you come to my party on Friday?" and Juan responded "I have to work", which means no.

Table 11: Performance on the implicature task framed contrastively by OpenAI's text-davinci-002. The mean and standard deviation are reported over two different prompt templates (template 1 and 2).

k	Non-contrastive	Rank one, two	Rank A, B	Rank full text
0	71.3% ± 1.75	53.9% ± 0.9	59.3% ± 1.3	48.9% ± 0.6
1	76.1% ± 2.6	59.4% ± 1.6	63.2% ± 2.0	66.9% ± 0.9
5	80.5% ± 2.3	61.4% ± 1.3	64.0% ± 1.3	67.9% ± 2.1

In Table 11, perhaps surprisingly, we can see that the contrastive task is much more difficult than the original ranking task. For $k = 0$, the result is random except for the prompt where the multiple choice options are A and B. For $k = \{1, 5\}$ the full text ranking does best, but is still significantly worse than the original ranking setup. Because of these disappointing results, we did not evaluate the other models contrastively. Future work must establish whether the contrastive setup is worse across all model classes and sizes.

K.2 Variance over prompt ordering

As mentioned in Section 3 in the main text, models are sensitive to the ordering of the k examples in the prompt. Instead of marginalising over this random factor by evaluating all possible prompt orderings, we randomly sampled an ordered set of examples from the development set for each test example. Throughout experiments, we kept this randomly sampled order the same, meaning if you re-run the 5-shot evaluation you get exactly the same orderings. The reason for this is that we want evaluate each model equally. In this section we ask how the performance chances for the best performing model if we select another random order. We do this for the 5-shot evaluation, because the results show that adding more in-context examples barely helps performance.

Table 12: Variance over prompt ordering for 5-shot evaluation per prompt template (P.T.) for text-davinci-002

Seed	P. T. 1	P. T. 2	P. T. 3	P. T. 4	P. T. 5	P. T. 6	Mean
0	80.17	78.17	82.83	80.50	79.17	76.50	79.56
1	80.17	76.17	81.33	81.83	76.00	76.33	78.64
2	79.50	78.17	81.17	80.17	78.17	76.50	78.94
mean	79.94	77.50	81.78	80.83	77.78	76.44	-
std	0.31	0.94	0.75	0.72	1.32	0.08	-

Table 12 shows the results of this experiment. Some prompt templates seem to be more sensitive to prompt example ordering than others, but for none of them the variance is high enough to change any conclusions.

K.3 Different zero-shot instruction prompts

There is a narrative around large language models that if they fail a task, it might be that the prompt was not the right one (through works like Reynolds and McDonell (2021b); Kojima et al. (2022)). The idea is that they can be prompted to simulate almost anything, if you set them up correctly. Because implicature resolution is a ubiquitous result of learning language, we hold the view that a model should be able to do this task if a prompt is given in coherent natural language. Nonetheless, in an additional effort to find the "let's think step-by-step" (Kojima et al., 2022) of zero-shot implicature resolution we try three more prompt templates. We evaluate a base large language model and two instructable models: GPT-3-175B, text-davinci-001, and text-davinci-002. The prompts we use are taken from recent work that proposes a dialogue agent trained with human feedback (Glaese et al., 2022), but adapted to the task of implicature resolution. The full prompts are presented in Table 7 and Table 13 shows the results. The new templates do not improve performance for any of these models. The variance over the prompt templates for text-davinci-002 is high, and the best prompt template of these three does achieve a slightly higher accuracy than the others: 74.5%. These results do not change the picture sketched so far.

Table 13: Zero-shot accuracy over three additional prompt templates for a base LLM and two instructable models.

Model	Templates
GPT-3-175b	59.2% $\pm$ 4.5
text-davinci-001-?	66.1% $\pm$ 3.2
text-davinci-002-?	67.7% $\pm$ 9.6

K.4 The effect of in-context examples on sensitivity to prompt wording

Figure 7 shows the relative performance increase due to in-context prompting broken down per prompt template. For text-davinci-001, most templates benefit similarly from more in-context examples, except for template 1. Perhaps surprisingly, we see that this template already achieves a performance of 76.5% at the zero-shot evaluation and does not improve much with few-shot prompting. For Cohere-52B and OPT-175B we see a clear grouping between the structured prompts (dashed lines) and natural prompts (dotted lines). Cohere struggles significantly more with the structured prompts than with the natural prompts in the zero-shot evaluation, and few-shot prompting can mitigate that, lowering the standard deviation over prompt templates to 1.89 at $k = 30$ from 4 at $k = 0$. OPT benefits from prompting for the natural prompts, but not for the structured prompts.